Implementation of Data Integrity Constraints

Primary keys, foreign keys, unique, check, and default rules

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Project: Library Management System. **Output convention:** result tables below are reproducible expected outputs from the fixed rows in the shared SQL script. They are not screenshots of MySQL execution.

1. Integrity constraints

Constraints make invalid relationships and values fail close to the data. This schema uses primary keys, foreign keys, unique identifiers, checks, and defaults. MySQL 8.0.16 or later is required for enforced CHECK constraints.

Constraint	Where used	Purpose
PRIMARY KEY	Every table	Prevents duplicate or missing row identity.
FOREIGN KEY	Books, junctions, copies, loans, reservations	Requires each reference to point to an existing parent row.
UNIQUE	ISBN, publisher/category name, member number/email, accession code	Prevents duplicate business identifiers.
CHECK	Price, publication year, loan dates, status and role	Rejects values outside the rules encoded in the table definition.
DEFAULT	Member join date, copy status, reservation status	Provides an explicit initial value when the insert omits that column.
NOT NULL	Required identity and descriptive fields	Prevents incomplete parent and transaction rows.

2. Successful verification queries

These checks should return zero invalid or orphan rows after the shared seed data has been inserted.

```
SELECT COUNT(*) AS orphan_book_rows
FROM books AS b
LEFT JOIN publishers AS p ON p.publisher_id = b.publisher_id
LEFT JOIN categories AS c ON c.category_id = b.category_id
WHERE p.publisher_id IS NULL OR c.category_id IS NULL;

SELECT COUNT(*) AS orphan_loan_rows
FROM loans AS l
LEFT JOIN book_copies AS bc ON bc.copy_id = l.copy_id
LEFT JOIN members AS m ON m.member_id = l.member_id
WHERE bc.copy_id IS NULL OR m.member_id IS NULL;

SELECT COUNT(*) AS invalid_price_rows
FROM books
WHERE list_price < 0;

SELECT COUNT(*) AS duplicate_isbn_groups
FROM (
    SELECT isbn FROM books GROUP BY isbn HAVING COUNT(*) > 1
) AS duplicate_isbns;

SHOW CREATE TABLE members;
```

Expected verification output

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

Check	Expected result
orphan_book_rows	0
orphan_loan_rows	0
invalid_price_rows	0
duplicate_isbn_groups	0
SHOW CREATE TABLE members	Displays PK member_id, unique member_no/email, and joined_on default

3. Constraint rejection examples

Run each statement separately only if you want to capture a rejection message. Each is expected to fail and is intentionally excluded from the shared setup script.

```
-- Expected duplicate email rejection (unique constraint)
INSERT INTO members
    (member_id, member_no, full_name, email, phone, joined_on)
VALUES
    (99, 'LIB099', 'Duplicate Email Test',
     'asha.mehta@example.edu', '+91-90000-00099', '2026-10-23');

-- Expected CHECK rejection (negative price)
INSERT INTO books
    (book_id, isbn, title, publication_year, publisher_id, category_id, list_price)
VALUES
    (99, '9780000000000', 'Invalid Price Demo', 2025, 1, 1, -1.00);

-- Expected FOREIGN KEY rejection (unknown publisher_id)
INSERT INTO books
    (book_id, isbn, title, publication_year, publisher_id, category_id, list_price)
VALUES
    (98, '9780000000001', 'Invalid Publisher Demo', 2025, 999, 1, 100.00);
```

4. Demonstrating a default without changing the seed data

```
START TRANSACTION;
INSERT INTO reservations
    (reservation_id, member_id, book_id, reservation_date)
VALUES (99, 5, 8, '2026-10-23');
SELECT reservation_id, reservation_status
FROM reservations
WHERE reservation_id = 99;
ROLLBACK;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

reservation_id	reservation_status
99	PENDING

ROLLBACK removes the demonstration row, leaving the four Assignment 7 reservations unchanged. Capture a successful verification query and, if required, one isolated rejection in MySQL Workbench.